DSA-210 Project Proposal

Motivation

In recent years I have been interested in Formula 1 and have become a big fan of sport. To win the races, teams need to analyze and adapt to changing conditions like weather, durations of pit stops and tracks. The aim of this project is to understand the effects of these parameters on winning races in Formula 1.

Data Collection

In Kaggle **this** dataset contains the results of every race and duration of pitstops in each race between the seasons 1950 and 2025. These are stored in different .csv files and these will be merged. Furthermore, this dataset will be enriched by obtaining the weather conditions for each race day through **Meteostat**. Weather conditions, like temperature and wind speed, of the racetrack during that day, and time can be accessed using the coordinates of the racetrack. Meteostat is an open-source project and has a python library to request this data. Lastly, the dataset will be enriched by the information of tracks which will be collected from **this** Kaggle dataset. It contains information about each track like number of turns, the length and the type.